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Research paper

Hybrid path planning framework to integrate improved A*-DWA algorithms for enhancing path safety and efficiency

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ABSTRACT

As the deployment of Unmanned Surface Vehicles (USVs) expands across various sectors, the importance of safe and efficient path planning systems is increasingly underscored, particularly in fields such as maritime safety and national security where high levels of navigational efficiency and safety are demanded. While previous path planning research often focused on either safety or efficiency, this study proposes an improved path planning framework that considers both aspects simultaneously, overcoming the limitations of traditional Global Path Planning (GPP) and Local Path Planning (LPP) to enhance route safety, efficiency capabilities. The proposed framework comprises GPP with Safety Areas (GPP-SA) and LPP with Goals & Obstacles (LPP-GO). The proposed structure ensures that the USV can navigate safely and efficiently in a variety of environments and reach its destination easily. The framework has been evaluated through various experiments, including scenario-based validation and statistical verification, demonstrating its superiority by integrating and improving upon existing methods. The findings of this research explore the feasibility of applying these advancements in real maritime environments, making a significant contribution to the realization of safe and efficient path planning in unmanned maritime and ship navigation systems.

1. Introduction

In recent years, the operation of unmanned surface vehicles (USVs) has gained increasing importance across various fields (Sarda and Dhanak, 2016; Barrera et al., 2021; Wang et al., 2019; Guo et al., 2019). USVs have become essential tools for exploring inaccessible areas and continuously collecting information, particularly during missions such as search, surveillance, and reconnaissance (Yuan and Rui, 2023; Dong et al., 2015; Zhao et al., 2021; Wang et al., 2019). As the application of USVs expands in areas such as maritime safety and national security, the importance of navigation systems for efficient and safe operation has become critical. In particular, safe and efficient path planning is a key component of navigation systems and essential for USVs to successfully accomplish missions (Specht et al., 2017; Yu et al., 2021; Tanakitkorn, 2019; Liu et al., 2016). For example, USVs in search and rescue missions during disasters rely on navigation systems to quickly navigate hazardous areas, avoid obstacles, and take optimal routes. This allows USVs to explore a larger area in a limited amount of time, significantly increasing the success rate of rescue efforts.

Fig. 1 shows the configuration of a typical path planning system for USVs, which is generally categorized into Global Path Planning (GPP) and Local Path Planning (LPP) (Bai et al., 2022; Chen et al., 2021; Yu et al., 2013; Sang et al., 2021). GPP involves establishing an overall optimal path from the origin to the destination, usually over a wide area. By contrast, the LPP follows the path set by the GPP while recognizing real-time changes in the environment (e.g., dynamic obstacles) and making detailed adjustments to avoid obstacles.

Various studies have been conducted based on this structure to enhance either the safety or efficiency of path planners. Voronoi diagrams (Niu et al., 2019) and grid-based methods (Song et al., 2019; Yang et al., 2017) have improved the safety of routes but have shown limitations in avoiding dynamic obstacles. Velocity obstacle methods (Wang et al., 2022) and Dynamic Window Approaches (DWA) (Luo et al., 2023) have enhanced real-time routing and flexibility but have struggled to achieve efficient path planning. Additionally, studies that apply the International Regulations for Preventing Collisions at Sea (COLREGs)

Abbreviations: USV, unmanned surface vehicles; GPP, global Path Planning; LPP, local path planning; GPP-SA, global path planning with safety area; LPP-GO, local path planning goal & obstacles.

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(Allen, 2021; Vagale et al., 2021) have demonstrated limitations in route optimization due to the need for strict rule compliance. These techniques have focused on one aspect and, as a result, have shown limitations in meeting complex requirements. This emphasizes the need for further research that considers both safety and efficiency in route planning to effectively navigate complex environments.

In this study, we propose a path planning framework that integrates enhanced GPP and LPP algorithms to simultaneously improve the safety and efficiency of unmanned surface vehicle (USV) navigation. This approach overcomes the limitations of previous methods that typically focused on only one aspect, ensuring robust operation in complex marine environments. From a safety perspective, the enhanced GPP employs an advanced A* algorithm augmented with safety buffers around static obstacles, enabling the USV to maintain a secure distance while determining the optimal route. This guarantees safe navigation even in areas densely populated with static obstacles. Moreover, the enhanced LPP utilizes an improved Dynamic Window Approach (DWA) to accurately predict the movements of dynamic obstacles, thereby facilitating real-time decision-making and effective collision avoidance. Overall, this integrated strategy addresses the need for a path planning system that simultaneously considers both safety and efficiency in dynamic environments.

From an efficiency perspective, the GPP framework has been developed by integrating Line of Sight (LOS)-based waypoint extraction directly into the A* algorithm. This integrated approach provides optimized routes directly without the need for any post-processing, significantly reducing the complexity of path planning and enhancing navigational efficiency. In the LPP, destination information is applied within the DWA, enabling the USV to efficiently navigate towards its target while effectively avoiding dynamic obstacles. This innovative strategy ensures that USVs maintain optimal paths, guaranteeing navigation stability and time efficiency in dynamic maritime environments.

To evaluate the safety and efficiency of the proposed framework, experiments were designed using safety metrics such as distance from obstacles and collision counts, along with efficiency metrics including path length, navigation time, USV turning frequency, and speed. In the first experiment, we tested the individual performances of the proposed GPP and LPP to verify their behaviors. In this experiment, we measured the obstacle avoidance performance and path efficiency to verify that each path planning method works as intended.

The second and third experiments involved simulation-based statistical evaluations of the integrated GPP and LPP frameworks. The second experiment was conducted in a small-scale map environment, and the third in a large-scale map environment. In both experiments, simulations were used to comprehensively assess the performance of the framework across different environments. The evaluation metrics used in these experiments included navigation time, planned path, and distance from obstacles, which were gathered for statistical analysis. Through these experiments, we compared the efficiency of path planning, obstacle avoidance capabilities, and overall navigation performance, thereby confirming the consistent performance of the framework in both scenarios.

The proposed path planning framework demonstrated significant improvements compared to conventional path planning frameworks. According to the experimental results, this framework reduced the average travel time by 60 % and shortened the path length by 15 %. Particularly in large-scale map environments, the framework consistently maintained a safe distance of >30 meters from both dynamic and static obstacles, proving its high safety levels. These results confirm that the proposed framework enhanced both the efficiency of path planning and obstacle avoidance performance, even in complex environments.

The remainder of this paper is organized as follows: Section 2 summarizes related studies based on path planning algorithms. Section 3 describes the integrated structure and algorithms of the proposed framework. Section 4 describes the experiments and results. Finally, our conclusions are presented in Section 5.

2. Related work

Table 1 summarizes previous research that proposed methods for

Table 1
Related work on path planning.

Reference	Static Obstacle Distance Consideration	Dynamic Obstacle Avoidance	Waypoint Simplification	Path Length Minimization
Niu et al. 2019	Yes	No	No	Yes
Song et al. (2019)	No	No	Yes	Yes
Yang et al. (2017)	Yes	Yes	No	No
Wang et al. (2022)	Yes	Yes	No	No
Luo et al. (2023)	No	Yes	No	No
Panov et al. (2018)	No	Yes	No	No
This paper	Yes	Yes	Yes	Yes

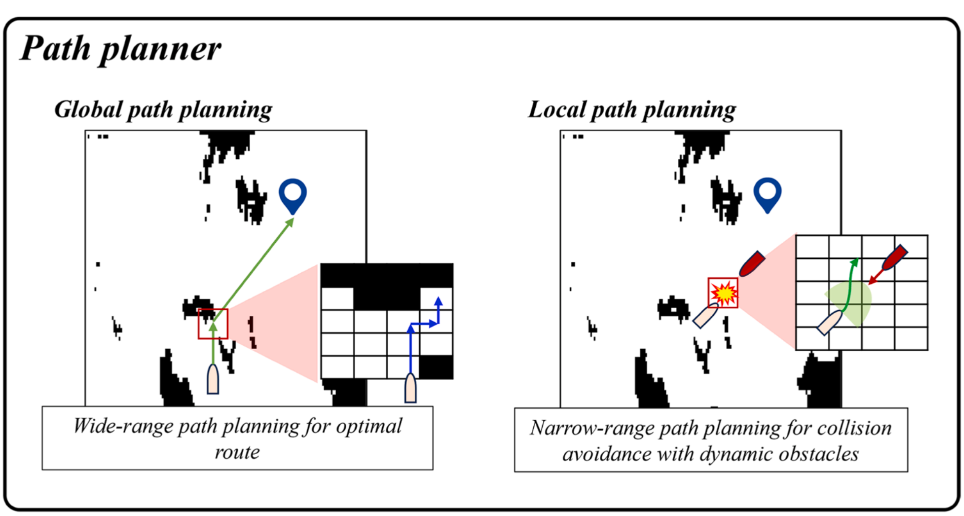

Fig. 1. Configuration of the typical path planner.

safe or efficient path planning. It highlights the key features of each path planning method, with a focus on safety, efficiency, obstacle avoidance, and waypoint simplification. Building on this research, our proposed method aims to develop a reliable and optimized path planning solution tailored for navigating USVs in complex marine environments.

Niu et al. (2019) utilized a Voronoi graph-based path planner (Zhao et al., 2022; Schoener et al., 2022) to ensure a minimum distance from obstacles, enhancing navigational safety. However, its lack of dynamic obstacle avoidance and high computational complexity limited its real-time applicability. Recent hybrid Voronoi-Grid approaches have integrated real-time dynamic re-routing algorithms that prioritize computationally efficient recalculation of local pathways when encountering sudden obstacles. These algorithms employ incremental graph updates instead of full recalculations, significantly reducing computational overhead while maintaining reliable obstacle clearance.

Song et al. (2019) implemented a grid-based planner with enhanced A* (Guo et al., 2022; Bertaska et al., 2015; Mousazadeh et al., 2018; Zhang et al., 2024), introducing waypoint simplification to optimize path efficiency. However, its inability to preemptively address dynamic obstacles resulted in frequent reactive adjustments, causing delays. Recent advancements use trajectory prediction models, integrating Kalman Filters or Bayesian Predictive Techniques to estimate obstacle motion over short time horizons. This enables the system to proactively adjust waypoints before obstacles intersect critical path segments.

Yang et al. (2017) utilized a cell decomposition-based path planner (Liang et al., 2021; Shriyam et al., 2018; Du et al., 2019; Zhang et al., 2024) for USV path planning, with the advantage of enabling dynamic obstacle avoidance and adaptive re-planning. This approach enables more effective obstacle avoidance even in complex marine environments. However, the planner did not implement waypoint simplification or path-length minimization, thus limiting its overall efficiency. To address these issues, recent research has introduced spatial clustering algorithms such as DBSCAN(Density-based spatial clustering of applications with noise) and K-Means to optimize path distribution by dynamically adjusting the density of waypoints in obstacle-dense areas (Gao et al., 2022).

Wang et al. (2022) focused on dynamic obstacle avoidance for USVs by using an improved Velocity Obstacle Method (Tang, 2023; Mousazadeh et al., 2018). The algorithm effectively integrates real-time environmental changes and obstacle data, allowing USVs to avoid dynamic obstacles more efficiently. This approach significantly improves navigational safety and obstacle avoidance performance in complex maritime environments. However, this method does not incorporate waypoint simplification or path length minimization, which limits the overall path efficiency. Recently, heuristic-guided hybrid VOM algorithms that incorporate techniques such as A* and genetic algorithms (GAs) have been studied.

Luo et al. (2023) applied an enhanced Dynamic Window Approach (DWA) (Liu et al., 2019; Zhang and Chen, 2021; Han et al., 2022) for path planning in dynamic environments to improve trajectory evaluation and ensure operational safety. The method excelled in obstacle avoidance and path flexibility, but did not incorporate waypoint simplification or path length minimization, which limited its performance in creating optimal routes. Particularly, the planning process is complex in dynamic environments with unexpected obstacles. To address the challenges, recent research has focused on dynamically adjusting path parameters, balancing obstacle avoidance with path efficiency, and reducing computational complexity.

In recent years, deep-learning-based reinforcement learning has garnered significant attention in the field of path planning. Panov et al. (2018) proposed a deep Q-learning approach designed to enhance traditional rule-based path planning algorithms. This method demonstrated notable improvements, particularly in dynamic environments with small-scale maps and irregular obstacles.

However, there are still difficulties in generating optimal routes in certain scenarios, and performance is still uncertain in novel or unseen environments. In addition, mismatches between the training and deployment environments can lead to unexpected problems that affect the adaptability and robustness of the algorithm. Therefore, deep learning-based path planning algorithms are not yet considered mature enough to be reliably deployed in real-world applications.

Recent path planning research has focused on partial recalculation, predictive modeling, waypoint density adjustment, dynamic speed control, and multi-criteria decision analysis. However, the trade-off between scalability and real-time adaptability remains a major challenge, highlighting the need for an integrated approach to address this challenge.

In this study, our proposed path planning framework adopts a unique approach by simultaneously considering both safety and efficiency. While many existing studies tend to focus primarily on one aspect, our framework integrates these two crucial factors, aiming to enhance them concurrently. This approach seeks to ensure safety while maximizing the efficiency of routes in maritime environments for USVs, a combination that has been relatively underexplored in the field. Thus, our research opens new horizons by addressing both of these key elements together, contributing to the advancement of maritime navigation systems. This dual focus is particularly innovative, offering substantial improvements that could redefine standards for unmanned surface vehicle operations in challenging marine conditions.

3. Proposed path planning framework

This section describes the detailed modeling of the global and local path planners utilized in the proposed path planning framework. In addition, we compared the characteristics of the proposed path planning algorithms with those of classical path planning algorithms.

3.1. Overall framework design

Fig. 2 illustrates the configuration of the proposed path planning framework, comprising GPP-SA (Global Path Planner with Safety Area) and LPP-GO (Local Path Planner with Goal & Obstacle velocity). GPP-SA converts the original map into a grayscale image (Huang et al., 2012; Qu and Zhang, 2010), then into a binary format, defines a safety area based on the LOS (Liu et al., 2016) and employs the A* algorithm (Song et al., 2019) to propose safe pathways that avoid terrain. In contrast, LPP-GO plans paths by considering the goal and velocity of the dynamic obstacles. It utilizes the DWA (Luo et al., 2023) to adjust the path in real time, effectively navigating dynamic obstacles. Both planners are designed to ensure the safe and efficient navigation of vessels, focusing on real-time adaptability and safety in complex environments.

The proposed framework is developed to guarantee the safety and efficiency of vessels in dynamic and unpredictable environments. To address the inability of traditional navigation systems to effectively adjust and respond to unexpected obstacles or environmental changes in real-time, advanced algorithms have been incorporated to help ships detect immediate threats and strategically plan their routes. This dual approach minimizes fixed and moving risks, increases operational safety, and reduces the possibility of accidents at sea. By focusing on global and local path planning, this design provides a comprehensive solution that can flexibly respond to a wide range of navigational challenges.

Fig. 3 depicts the structured process of the proposed path planning framework, which is organized into two principal phases for effective maritime navigation. The initial phase begins with the collection of global information, which is crucial for initializing the GPP-SA. This stage involves setting the starting and destination waypoints essential for defining the overall navigation route. The system then evaluates whether the collected data, including maps and area-detection features, are adequate for path planning. Upon confirming sufficient information, a safety area is established to enhance navigation safety within which the GPP-SA operates. The GPP-SA performs LOS-based waypoint

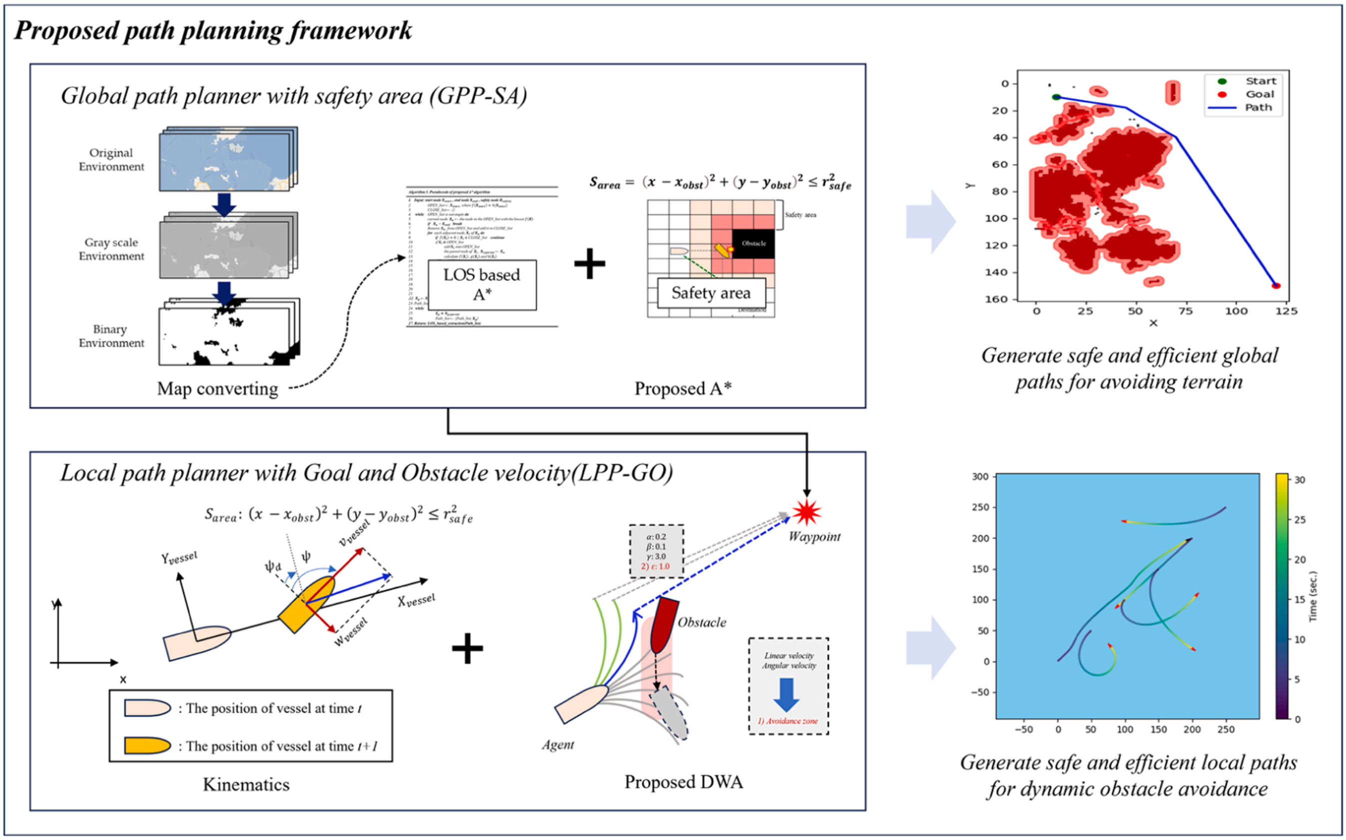

Fig. 2. Configuration of the proposed path planning framework.

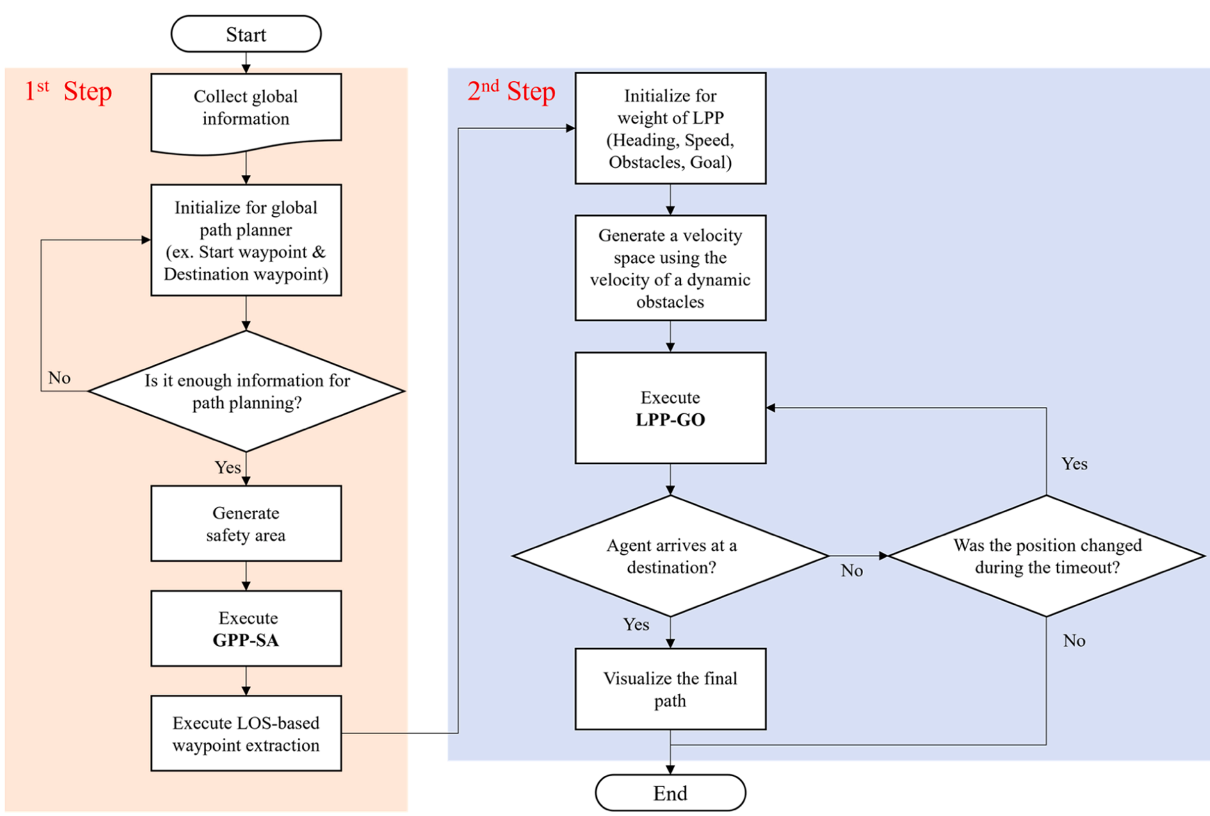

Fig. 3. Operational flowchart for proposed path planning framework.

extraction to finalize the global path, ensuring that the route is optimized for both safety and efficiency.

The next step involves executing the LPP-GO, which starts by setting the initial parameters, such as the vessel's current direction, velocity, information about dynamic obstacles in the vicinity, and the ultimate destination. These parameters are instrumental in forming a velocity space that accounts for the motion of dynamic obstacles and the destination, thereby facilitating effective real-time path adjustments. The LPP-GO algorithm is executed iteratively, adapting the path as the USV moves towards its destination. Upon arrival, the system visualizes the final path and offers a clear and practical representation of the traveled path.

The two-step framework provides a logical and organized sequence of tasks, enhancing the efficiency of path planning in complex maritime environments. Each phase is designed to complement the other phases and ensure that the vessel navigates safely and efficiently by integrating comprehensive global data with precise local adjustments.

3.2. Global path planner with safety area

Fig. 4 compares the classic A* algorithm used in GPP and the enhanced A* algorithm used in GPP-SA for the same scenario. The upper illustration offers a detailed side-by-side comparison of both path planning methods at specific times, clearly illustrating the positional changes and routes of the USV, and how each method adapts to obstacles.

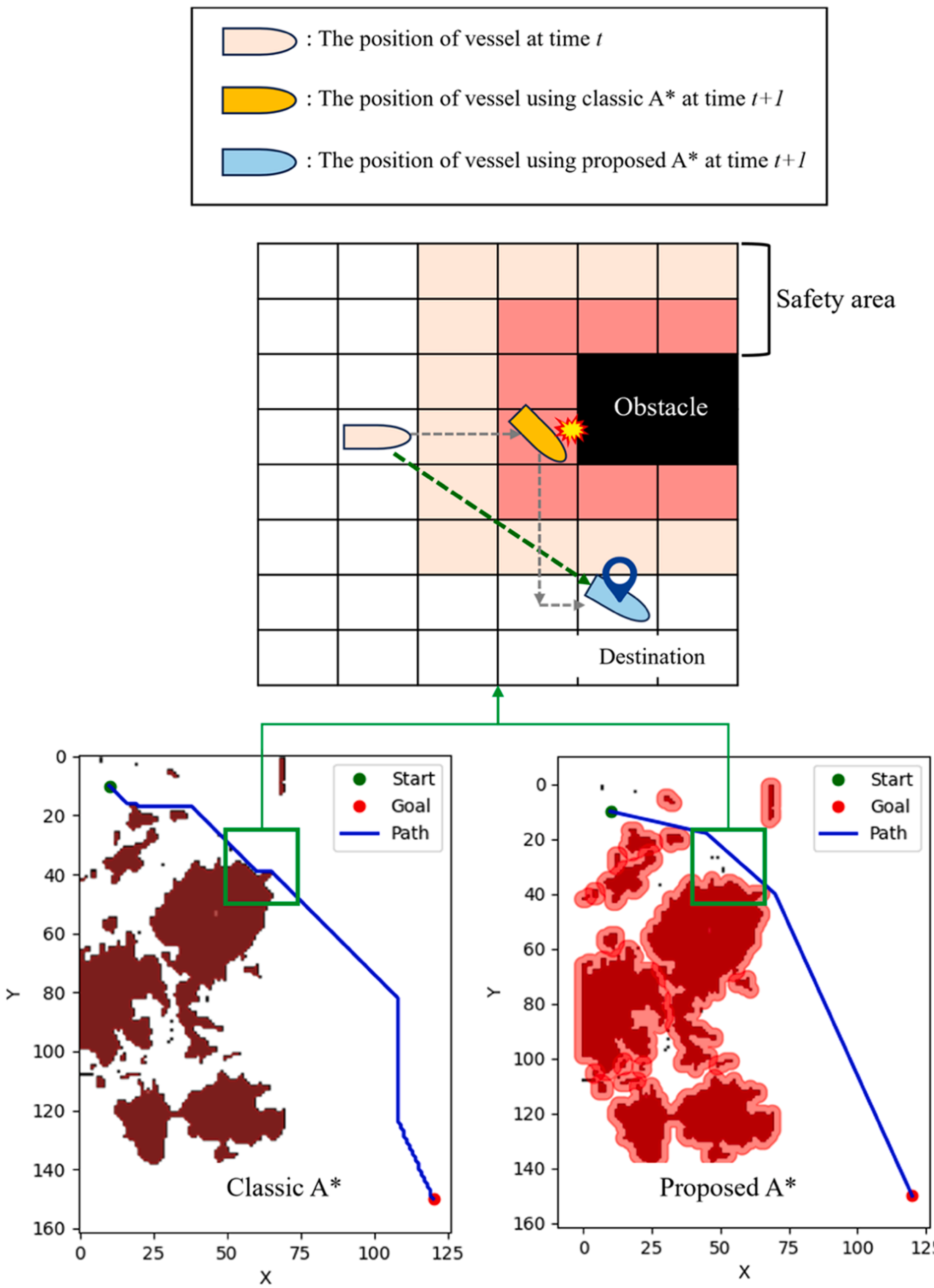

Fig. 4. Differences between classic A* and proposed A* algorithms.

The lower illustration in Fig. 4 visually compares the paths generated by the two algorithms under actual terrain conditions. Both the classic GPP and enhanced GPP-SA start from the same point and are directed towards the same destination. However, at time $t + 1$, the routes generated by GPP-SA and classic GPP were different, with GPP-SA providing a safer and more efficient route by generating safety areas. It selects paths that minimize the risk of collision with obstacles while reducing the distance to the destination, thereby improving the navigational efficiency and safety of the USV.

The enhanced A* algorithm in GPP-SA demonstrates superior path planning capabilities compared with the classic methods used in GPP. By employing safety areas and LOS-based waypoint extraction for streamlined path planning, the GPP-SA effectively circumvents unforeseen obstacles in maritime environments, ensuring that the USV reaches its destination more swiftly and securely. This represents a significant technological advancement in navigating under complex maritime conditions.

3.2.1. Safety buffer generation

Algorithm 1 shows the pseudocode for generating a safety buffer in the GPP-SA. The algorithm first takes as input a list of obstacles, safety distance, and safety cost; then generates a safety region based on the safety distance defined around each obstacle; and finally outputs the cost assigned to each node in the region. The assigned cost quantifies the risk, and path planning uses this information to help select a route that minimizes risk.

The mathematical expression for creating a safety buffer is given in Eq. (1): This was used to calculate the circular safety area around each terrain type. Eq. (1) is used to calculate the circular safety areas around each obstacle with a radius of r_{safe}, where x_{obst}, y_{obst} represent the coordinates of the obstacle. All the nodes within the circle are assigned a high safety cost, which instructs the path planner to avoid this area.

Algorithm 1

Pseudocode of safety area generation.

Input: obstacles, safety_distance, safety_cost
Output: safety_area with assigned costs
1. *safety_areas ← {}*
2. *for each obstacle in obstacles do*
3. *Calculate safety_area around obstacle with radius safety_distance*
4. *Assign safety_cost to each node within safety_area*
5. *safety_areas[obstacle] ← safety_area*
6. *Return safety_areas*

Algorithm 2

Pseudocode of proposed A* algorithm.

Input: $\boldsymbol{X_{start}}$, $\boldsymbol{X_{end}}$, $\boldsymbol{S_{area}}$
Output: Path_list
1. *OPEN_list ←* $\{\boldsymbol{X_{start}}\}$
2. *CLOSE_list ← {}*
3. *while OPEN_list is not empty do*
4. $\boldsymbol{X_n}$ *← the node in the OPEN_list with the lowest total cost*
5. *if* $\boldsymbol{X_n} = \boldsymbol{X_{end}}$ *break*
6. Move$\boldsymbol{X_n}$ *from OPEN_list to CLOSE_list*
7. *for each adjacent node,* $\boldsymbol{X_i}$ *of* $\boldsymbol{X_n}$ *do*
8. *if* $\boldsymbol{X_i} \in$ *CLOSE_list, continue*
9. *if* $\boldsymbol{X_i} \notin \boldsymbol{S_{area}}$ *then*
10. $\boldsymbol{g}(\boldsymbol{X_i}) \leftarrow \boldsymbol{g}(\boldsymbol{X_i}) + \boldsymbol{g}(\boldsymbol{S_{area}})$
11. *if* $(\boldsymbol{X_i})$ *already in OPEN_list and* $\boldsymbol{g}(\boldsymbol{X_i}) \geq \boldsymbol{g}(\boldsymbol{X_n})$ *continue*
12 *end if*
13. $\boldsymbol{g}(\boldsymbol{X_i}) \leftarrow \boldsymbol{g}(\boldsymbol{X_i}) + \boldsymbol{d}(\boldsymbol{X_n}, \boldsymbol{X_i})$
14. $\mathrm{h}(\boldsymbol{X_i})$ *← heuristic distance from* $\boldsymbol{X_i}$*to* $\boldsymbol{X_{end}}$
15. $f(\boldsymbol{X_i}) \leftarrow \mathrm{g}(\boldsymbol{X_i}) + \mathrm{g}(\boldsymbol{S_{area}}) + \mathrm{h}(\boldsymbol{X_i})$
16 *Add* $\boldsymbol{X_i}$ *to OPEN_list if not already present*
17 *end for*
18 *Sort OPEN_list by f(X) values*
19. *end while*
20. *Path_list ← Backtrack from* $\boldsymbol{X_{end}}$*to* $\boldsymbol{S_{area}}$ *using parent nodes*
21. *Return: LOS_based_extraction(Path_list)*

$$S_{area} : (x - x_{obst})^2 + (y - y_{obst})^2 \leq r_{safe}^2 \tag{1}$$

Using Eq. (1), GPP-SA enables vessels to safely reach their destinations in complex maritime environments. The safety area is an important factor in minimizing the risk of collisions with obstacles and improving the accuracy and reliability of path planning.

3.2.2. Path planning for GPP-SA

Algorithm 2 describes the process of determining a path from a starting point (X_{start}) to an ending point (X_{end}) using the enhanced A* algorithm. In contrast to the classic A* algorithm, the proposed algorithm incorporates a safety area (S_{area}), which ensures safe path planning by penalizing nodes located within the obstacle's defined boundary. The inputs to the algorithm include the starting point, end point, and obstacle list. The algorithm evaluates the total cost ($f(X)$) for each node, which consists of the actual cost ($g(X)$), a penalty for entering the safety area(($g(S_{area})$)), and a heuristic estimate of the remaining distance ($h(X)$).

By explicitly integrating S_{area}, the algorithm prioritizes nodes that maintain safe distances from obstacles, thereby minimizing collision risks and improving path safety. This proactive approach enables the proposed algorithm to generate more stable and efficient paths compared to traditional methods. Once X_{end} is reached, the final path is refined using an LOS-based extraction method to further optimize the route before it is passed to LPP-GO.

Algorithm 3 describes the process of extracting a route based on the LOS from a starting point to a goal point. This algorithm was used to plan safe and efficient routes in environments with obstacles. It aims to avoid obstacles and minimize route changes, maintaining the path close to a straight line.

The path is initialized from the starting point, and the current location is set as the starting point. The algorithm was executed iteratively until the current position reached the goal point. In each iteration, the next point was calculated from the current location towards the goal point, and the straight path from the current location to the next point was checked for intersections with obstacles. If the path intersects an obstacle, the LOS is considered blocked, and a detour is calculated considering the safe distance. If the line-of-sight was clear, the next point was added to the direct route; if it was blocked, a detour point was added to the route.

Algorithm 3 is important for determining the optimal route to the destination while effectively avoiding obstacles. LOS-based path extraction allows users to reach their destination quickly and safely even in complex environments with obstacles, minimizing the risk of collisions and contributing to overall navigational safety.

Algorithm 3

Pseudocode of LOS_based_extraction.

Input: start, goal, obstacles
Output: Path
1. *path ← start*
2. *current_point ← start*
3. *while current_point != goal do*
4. *next_point ← calculate_next_point(current_point, goal)*
5. *los_clear ← True*
6. *for each obstacle in obstacles do*
7. *if line_intersects_obstacle(current_point, next_point, obstacle) then*
8. *los_clear ← False*
9. *break*
10. *if los_clear then*
11. *path.append(next_point)*
12. *current_point ← next_point*
13. *else*
14. *detour_point ← calculate_detour(current_point, next_point, obstacles, safety_distance)*
15. *path.append(detour_point)*
16. *current_point ← detour_point*
17. *return path*

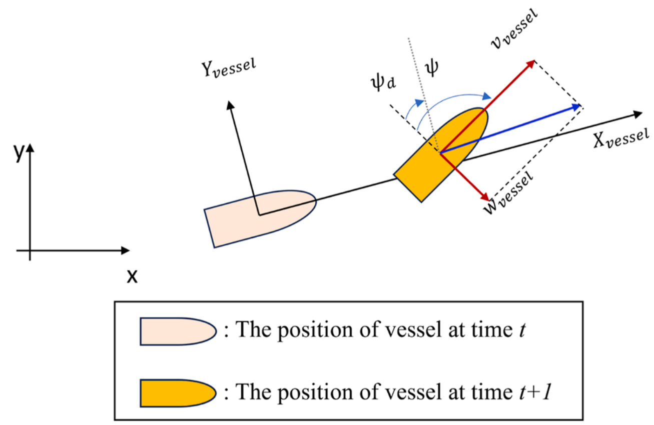

Fig. 5. The USV mathematical motion model.

3.3. Local path planner with goal & obstacle velocity

3.3.1. Kinematic model

Fig. 5 (Guan and Wang, 2023) illustrates the mathematical motion model of an USV, visualizing the changes in the USV's position and direction. In the diagram, X_{vessel} Y_{vessel} represent the USV's coordinates along the x-axis and y-axis, respectively, while ψ indicates the current direction, and ψ_d the desired direction. The terms ν_{vessel}, and ω_{vessel} represent the linear velocity and angular velocity, respectively, showing the speed of the USV's linear and rotational movements.

Eq. (2) (Liu et al., 2016) mathematically defines the dynamic motion model and provides a method for calculating the dynamic response of a USV. Here, r denotes the USV's current angular velocity, δ is the current steering angle, and δ_E is the target steering angle. K and T are coefficients that characterize the system, adjust the rate of change of the linear velocity, and aid the USV in reaching its target steering angle more stably. This mathematical approach ensures that the USV can effectively navigate along its intended path and avoid obstacles.

$$\begin{bmatrix} \dot{\psi}\dot{r}\dot{\delta} \end{bmatrix} = \begin{bmatrix} r \\ (K\delta - r)/T \\ \delta_E - \delta \end{bmatrix} \tag{2}$$

Consequently, Fig. 5 and Eq. (2) are important for modeling the dynamic behavior of the USV and formulating effective path planning and steering strategies. These visual and mathematical tools are essential for designing and simulating processes that maximize navigational safety and efficiency in complex marine environments.

Table 2 lists the dynamic parameters of the USV used in this study. It details the specifications essential for understanding and simulating the behavior of the USV under different conditions. Knowing and controlling these parameters is essential for the LPP to ensure safe and efficient operation of the USV in complex maritime environments, as it enables the USV to respond effectively to different sea conditions and dynamic obstacles and safely reach the target destination in an optimal path.

Eq. (3) is important for calculating changes in the position and direction of the USV. This provides an important basis for using the DWA algorithm, which is a sub-algorithm of LPP-GO. DWA uses linear (v_{vessel}) and angular (w_{vessel}) velocity to predict the position (x,y) and orientation (ψ) of the USV at the next time step and calculates the optimal path at each point in time.

$$\begin{cases} x(t+1) = x(t) + v_{vessel}\Delta t\cos(\psi(t)) - w_{vessel}\ \Delta t\ sin(\psi(t)) \\ y(t+1) = y(t) + v_{vessel}\Delta t\sin(\psi(t)) - w_{vessel}\ \Delta t\ cos(\psi(t)) \\ \psi(t+1) = \psi(t) + r\Delta t \end{cases} \tag{3}$$

Table 2
Dynamics parameters of experiment 1.

Parameter	Value	Description	Unit
Length	5	Length of the vessel	m
Max speed	20	Maximum speed in knots	knots
Max rudder angle	30	Maximum rudder angle in degrees	Degree
Max rudder angle rate	20	Maximum rate of change of rudder angle	Degree/s
K	−0.085	Stability constant for the vessel's dynamics	–
T(s)	4.2	Time constant related to the vessel's turning dynamics	s

The DWA algorithm considers the current position of the USV and possible movements to plan a safe and efficient trajectory. The predicted position and heading calculated using Eq. (3) are used by the DWA to dynamically adjust for changes in the angular velocity and line speed, avoid obstacles, and optimize the path to the target point in real time. This process provides a safe and reliable path for the USV to reach its target, while effectively avoiding obstacles in various maritime environments.

3.3.2. Path planning for LPP-GO

Fig. 6 illustrates the concept of LPP by showing two different DWA algorithms: classic DWA and the proposed DWA. Fig. 6(a) shows the classic DWA, which visualizes potential trajectories for the USV, whereas Fig. 6(b) shows the proposed DWA, which includes enhancements in trajectory evaluation and path optimization.

Specifically, Fig. 6(a) shows a classical DWA that calculates the probability of collision with an obstacle using the possible trajectories of the USV. However, these conventional approaches consider only the direction towards the destination and the current position of the obstacle to generate a path. Therefore, the probability of collision depends on the motion of the dynamic obstacle. In this context, dynamic obstacles are specifically represented by other navigating surface vessels, whose movements are predicted based on their prior speeds and directions. This prediction allows for a basic evaluation of collision risks, but conventional approaches often fail to account for more complex or nonlinear movements, leading to paths that may avoid obstacles but deviate significantly from the destination.

In contrast, Fig. 6(b) shows the proposed DWA, which considers not only the destination of the USV but also the expected motion of the dynamic obstacle based on its prior behavior. The proposed DWA is based on a combination of linear and angular velocities to evaluate and adjust the trajectory dynamically by considering the distance and direction of the destination. It also incorporates the velocity and the predicted path of the moving obstacle to proactively avoid potential collisions. This approach ensures that the USV chooses a safe and optimized path for obstacle avoidance, enabling efficient path generation even in complex situations.

To apply the proposed DWA, the essential constraints of the dynamic window must be set. These constraints define the current kinematic state of the USV and dynamic velocity range that enables real-time reactions to the surrounding environment. This allows the USV to avoid obstacles and plan an optimal path to the goal point. Eq. (4) sets the physical limits of the USV and defines the safe operational speeds and rotational ranges. These limits reflect the mechanical restrictions and performance capabilities, ensuring that the USV operates within its technical specifications.

$$V_m = \{(v, r) | v \in [v_{min}, v_{max}],\ r \in [r_{min}, r_{max}]\ \} \tag{4}$$

Eq. (5) dynamically defines the range within which the USV can adjust its speed in real time, considering the possible accelerations and decelerations from the current speed. This capability allows the USV to appropriately adjust its velocity in response to sudden environmental changes or operational demands. Eq. (6) ensures safe navigation by preventing collisions with obstacles. It calculates the maximum deceleration speed by considering the distance to obstacles, enabling the USV to safely reach its destination without collision.

$$V_d = \{(v,r) | v \in [v_c - \dot{v}_b \Delta t,\ v_c + \dot{v}_a \Delta t],\ r \in [r_c - \dot{r}_b \Delta t,\ r_c + \dot{r}_a \Delta t]\} \tag{5}$$

$$V_a = \left\{(v,r) | v \leq \sqrt{2 \cdot dist(v,r) \cdot \dot{v_a}}\ ,\ r \leq \sqrt{2 \cdot dist(v,r) \cdot \dot{r_a}}\ \right\} \tag{6}$$

Eq. (7) integrates all constraints to determine the final set of feasible velocities that the USV can select. This final set ensures that the USV satisfies the physical, dynamic, and safety constraints while choosing the optimal path to its destination. Eq. (8) serves as the central evaluation function in the proposed DWA and plays a crucial role in the trajectory selection for the USV by comprehensively considering various factors. $G(v,r)$ incorporates different trajectory options based on their alignment with the desired path to the destination and by evaluating aspects such as direction, distance, speed, and goal proximity to determine the optimal path.

$$V_r = V_r\ \cap\ V_d\ \cap\ V_a \tag{7}$$

$$G(v,r) = \ \alpha \cdot heading(v,r) + \beta \cdot dist(v,r) + \gamma \cdot velocity(v,r\ ) + \varepsilon \cdot goal(v,r) \tag{8}$$

The components of this function include $heading(v,r)$, which assesses the difference between the current heading of the USV and the ideal direction to the target, guiding the USV towards a direct approach to the destination. $dist(v,r)$ evaluates how well the USV maintains a safe distance from obstacles, thus ensuring collision avoidance and safe navigation. $velocity(v,r)$ checks whether the current speed of the USV is appropriate for the situation, aiding in energy optimization and maneuverability through efficient speed management. Finally, $goal(v,r)$ measures how effectively the USV can approach its target using the distance between goals.

Each of these components is weighted by coefficients α, β, γ, and ε, which were determined through an empirical tuning process based on extensive experimental iterations. These coefficients were adjusted to optimize the balance between safety, efficiency, smoothness, and adaptability in various navigation scenarios, ensuring robust performance across different operating conditions.

- α: Represents the heading weight towards the destination, ensuring the path remains consistently directed toward the target point while minimizing unnecessary deviations.
- β: Reflects the weight associated with the distance from obstacles, prioritizing safe clearance and reducing the risk of collisions during navigation.
- γ: Controls the speed adjustment weight, regulating the velocity of the agent to maintain smooth and stable motion throughout the path.
- ε: Controls the distance weight to the destination, emphasizing the reduction of the remaining travel distance while ensuring efficient progress toward the target.

Through empirical tuning, these coefficients were iteratively optimized to strike an effective balance among competing objectives, ensuring that the framework performs consistently under diverse maritime navigation conditions. This parameter optimization process significantly improved the evaluation capability of the framework, allowing the USV to select optimal routes that align with mission-specific priorities.

Algorithm 4 outlines the pseudocode for generating an avoidance zone by predicting the positions of the dynamic obstacles, which is essential for ensuring that the USV navigates safely. This algorithm uses the current position of the USV ($P^{USV}_{current}$), obstacle position (P_{obs}), and *prediction_time* as inputs for processing the data. The output is the avoidance zone, around which the USV must navigate.

Algorithm 4 begins by creating an empty avoidance zone and dividing the prediction time into equal intervals (dt). The number of steps is calculated by dividing the total prediction time by these time intervals. For each step, the algorithm predicts the position of the obstacle based on its current location and linear and angular velocities. Future positions are calculated by adjusting the obstacle's linear and angular velocities over the time step, and each predicted position is added to the *avoidance_zone.*

Algorithm 5 describes the proposed DWA for dynamically adjusting the trajectory of a USV in real time based on the destination and obstacle locations. The algorithm uses linear and angular velocities, v^{DWA} and w^{DWA}, to simulate the potential motion of the USV. This motion is projected to a future position (X^{USV}_i, Y^{USV}_i, φ^{USV}_i), allowing different trajectory options to be evaluated based on these predicted coordinates. However, the proposed DWA destination requires further evaluations. The Euclidean distance (H^{USV}_i) is calculated to evaluate the destinations. A cost function (C_i) is then applied to incorporate this distance and other navigational factors to determine the desirability of each trajectory. C_i is important for evaluating the tradeoff between proximity to obstacles and the optimal path to the destination.

Finally, the algorithm iterates through all possible velocities within the set dynamic window and calculates the cost for each. The trajectory

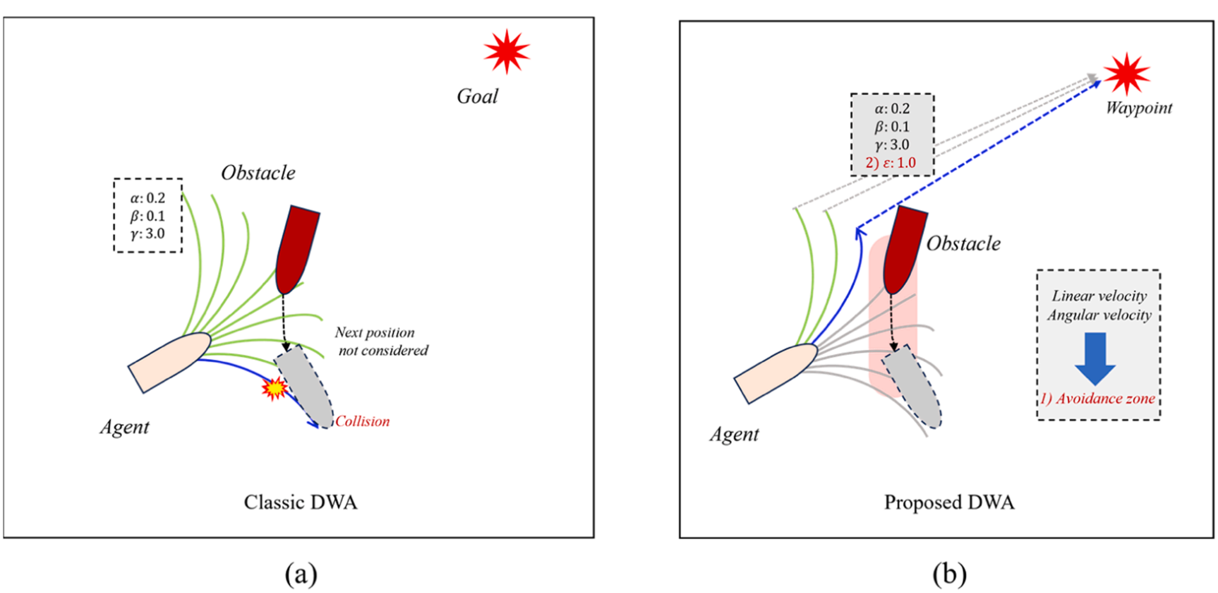

Fig. 6. Concep of proposed DWA algorithms. (a) Classic DWA. (b) Proposed DWA.

Algorithm 4
Pseudocode of avoidance zone generation.

Input: P^{USV}_{curent}, P_{obs}, *prediction_time*
Output: *avoidance_zone*
1. *avoidance_zone* ← {}
2. *time_step* ← *dt*
3. *num_steps* ← *prediction_time* / *time_step*
4. *zone_radius* ← *safety_margin*
5. *for step from 1 to num_steps do*
6. *predicted_position* ← P^{USV}_{curent}
7. *obstacle_position* ← P_{obs}
8. *predicted_position.x* += *obstacle_position.linear_velocity* * cos(*obstacle_position.angular_velocity*) * *time_step* * *step*
9. *predicted_position.y* += *obstacle_position.linear_vel`ocity* * sin(*obstacle_position.angular_velocity*) * *time_step* * *step*
10. Z_{step} ← *set(predicted_position.x, predicted_position.y)*
11. *avoidance_zone.add*(Z_{step})
12. *Return: avoidance_zone*

Algorithm 5
Pseudocode of proposed DWA.

Input: P^{USV}_{curent}, P_{obs}
Output: O^{USV}
1. *v_resolution* ← 0
2. *i* ← 0
3. *while* N^{v} > *v_resolution do*
4. $v^{DWA} \leftarrow v^{ini} + \Delta v(v\_resolution)$
5. *yaw_rate_resolution* ← 0
6. *while* N^{w} > *yaw_rate_resolution do*
7. $w^{DWA} \leftarrow w^{ini} + \Delta w(yaw\_rate\_resolution)$
8 $< X_i^{USV}, Y_i^{USV}, \varphi_i^{USV} > \leftarrow$ *motion*$(v^{DWA}, v^{DWA}, P^{USV}_{curent})$
9 $P_i^{USV} \leftarrow \langle X_i^{USV}, Y_i^{USV}, \Phi_i^{USV} \rangle$
10 $H_i^{USV} \leftarrow$ *euclidean_dist*(X_i^{USV}, Y_i^{USV})
11 $C_i \leftarrow$ *calculate_cost*$(P_i^{USV}, H_i^{USV}, P_{obs})$
12 *yaw_rate_resolution* ← *yaw_rate_resolution* + 1
13 *i* ← *i* + 1
14. *end while*
15 *v_resolution* ← *v_resolution* + 1
16. *end while*
17. $\langle x^{USV}, y^{USV}, \theta^{USV} \rangle \leftarrow \min(C, P^{USV})$
18. $O^{USV} \leftarrow \langle x^{USV}, y^{USV}, \theta^{USV} \rangle$
19. *Return:* O^{USV}

with the lowest cost is selected to present the most optimized and safest path for the USV. The selection process ensures that the USV can efficiently navigate complex environments, dynamically adapt to changes, and avoid collisions, while maintaining a safe and strategically feasible path.

4. Experimentation

4.1. Experimental design

This section details the experiments designed to validate the effectiveness of the proposed path planning framework. Table 3 provides the detailed configurations of these experiments, which were conducted using the proposed method and well-known conventional methods as controls. Three experiments were conducted to verify the operational functionality and performance, providing a clear comparative analysis of the framework's characteristics and excellence under various conditions.

The evaluation metrics were defined according to the purpose of each experiment. For Experiment 1, which focused on operational validation, the metrics included the visualization of paths and quantitative data to assess the effectiveness of the basic functions. For Experiments 2 and 3, which focused on performance evaluation, safety- and efficiency-related performance metrics were applied. These metrics reflect the statistical characteristics and systematically measure the generalized performance across various maritime conditions.

The experimental environments were designed to be divided between small and large maps and real and simulated environments, depending on the objective. Small-scale maps are used to assess the basic operations of the proposed framework algorithms, focusing on their ability to make rapid decisions and fine adjustments. Large-scale maps were used to test the performance of these algorithms under more extensive and complex conditions to evaluate the overall performance of the framework. Virtual environments facilitate the testing of algorithms under controlled conditions, enabling thorough assessment, whereas real environments validate the practical operation of the framework in actual maritime settings.

Experiment 1 on small-scale maps focused on evaluating the basic behavior of the algorithms used in the proposed framework and testing their responsiveness and decision-making abilities in a limited space, such as GPP-SA and LPP-GO. Experiments 2 and 3 aimed to evaluate the overall performance of the framework in different maritime environments by generalizing the performance of the algorithms tested on small-scale maps to larger and more complex environments, both on small and large-scale real-world maps. The validity and efficiency of the proposed path planning framework were demonstrated through these experiments, thereby confirming its suitability for real-world maritime operations. This comprehensive evaluation is crucial for assessing how well the proposed framework meets the demands of various maritime environments.

4.2. Experiment 1

4.2.1. Experiment 1(a): behavior validation using GPP-SA

Fig. 7 presents the results of Experiment 1(a). Fig. 7(a) depicts a real-world map environment measuring 1.1 km by 1.0 km, which includes

Table 3
Experimental scenarios for path planning.

Experimental Case	Objective	Evaluation metrics	Iterative simulation	Experimental environment	Experimental group (Proposed method)	Control group
Experiment 1(a, b, c)	Behavior validation	Visualization of a path plan, Quantitative metrics	X	Small-scale real map Small-scale virtual map Small-scale real map	GPP-SA LPP-GO GPP-SA & LPP-GO	Classic A* Classic DWA Classic A* & Classic DWA
Experiment 2	Performance evaluation	Safety and efficiency-related performance metrics	O	Small-scale real map	GPP-SA & LPP-GO	Classic A* & Classic DWA
Experiment 3		Safety and efficiency-related performance metrics	O	Large-scale real map	GPP-SA & LPP-GO	Classic A* & Classic DWA

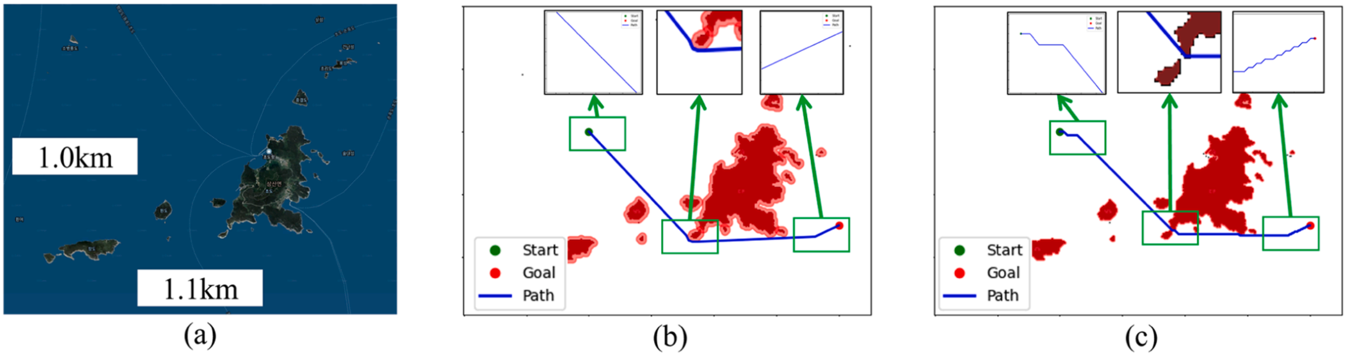

Fig. 7. Validation of the behavior of the GPP-SA. (a) Real-world map environments. (b) Proposed GPP-SA (c) Classic GPP.

complex terrains and water bodies, providing an ideal setting to test the navigation capabilities of the USV. Safety buffer areas were set approximately 5 m away from the terrain features and objects, considering the size of the USV. The buffer distance was selected based on the recommended practices to ensure safe navigation around obstacles. It can be adjusted as needed to provide a flexible and efficient approach for minimizing risks in maritime operations.

Fig. 7(b) shows the results obtained using the proposed GPP-SA. The inset map indicates that the path avoids light-red terrain areas, maintaining a safe margin while effectively navigating around terrain obstacles, thereby significantly enhancing the safety and efficiency of the USV in complex environments. Additionally, this path is created using LOS-based waypoint extraction, which substantially reduces the number of zigzag paths and creates more direct and efficient routes.

As depicted in Fig. 7(c), the results of using the Classic GPP reveal several critical issues. First, the algorithm generates paths that are dangerously close to obstacles, increasing the risk of collision and significantly compromising navigational safety. Second, due to the absence of LOS based waypoint extraction, the paths created zigzag, resulting in an unnecessarily high number of turning points. These zigzag paths reduce navigation efficiency and place additional load on the navigation system, making it difficult to maintain an optimal route.

These issues underscore the need for improvements in the path planning algorithm to enhance safety and efficiency in navigation. Addressing the shortcomings of Classic GPP and adopting enhanced path planning frameworks like GPP-SA are crucial for providing safer and more efficient navigational routes.

Table 4 illustrates that while GPP-SA did not significantly reduce travel distance compared to the basic A* algorithm, it effectively eliminated all collision risks and dramatically reduced the number of waypoints from 201 to just four through the use of safety areas. Moreover, this reduction in waypoints significantly decreased the load on the path following. The simplification of the path also drastically reduced the number of turning points from 22 to 3, minimizing the necessary adjustments during navigation and enabling a more stable and efficient route. This represents a critical improvement, significantly enhancing safety and overall efficiency in maritime environments with dynamic obstacles. GPP-SA, with these advantages, consistently outperforms traditional methods and effectively addresses the critical needs of dynamic maritime environments.

Table 4
Path planning results between GPP-SA and classic GPP.

	Travel distance (m)	Number of waypoints (pts)	Number of turning points(pts)
GPP-SA	956	4	3
Classic GPP	950	201	22

4.2.2. *Experiment 1(b): behavior validation using LPP-GO*

The proposed LPP-GO was validated using three primary scenarios to assess its efficiency and safety, as detailed in Fig. 8. Case 1 evaluated the capability of the USV to avoid a dynamic obstacle approaching from the front. The scenario in Case 2 required the USV to navigate between two dynamic obstacles approaching from the sides. Case 3 assessed the path planning capability in a complex environment with multiple dynamic obstacles. Fig. 8(a) displays the map environments in which obstacle avoidance experiments for each case were conducted. Figs. 8(b) and (c) illustrate the proposed LPP-GO and the route planned by the classic LPP, where the color of the route visually indicates the time spent and the red 'X' marks the destination.

Notably, in each case, the proposed LPP-GO tended to choose more direct and efficient paths by integrating the destination information and dynamic properties of obstacles, demonstrating superior performance in terms of time efficiency and safety. Conversely, in each scenario the LPP opted for longer and more circuitous routes, focusing solely on obstacle avoidance, which often led to less efficient paths.

A closer examination of Figs. 8 (3-b) and 8 (3-c) from Case 3 illustrates the distinctive advantages of the proposed LPP-GO over the classic LPP. Fig. 8 (3-b) shows the strategic and proactive path planning of the proposed LPP-GO, which successfully navigates through a complex environment with fewer turns and a more direct route, significantly reducing the travel time. The heat map of this trajectory indicates shorter final simulation times, demonstrating the efficiency of the proposed system in minimizing navigation time and enhancing safety by avoiding close encounters with obstacles.

By contrast, the path shown in Fig. 8 (3-c) emphasizes the reactive strategy of the classic LPP, which requires multiple adjustments and prolongs the navigation time owing to clusters of moving obstacles. The colors in the heatmap indicate the inefficiency and potential delay of the traditional path planning methods, highlighting the extended travel time. This contrast clearly demonstrates the ability of the LPP-GO to integrate dynamic obstacle movements and destination considerations to deliver significant time efficiency and navigational safety benefits in complex maritime environments.

Table 5 presents the quantitative results of the dynamic obstacle-avoidance experiment depicted in Fig. 8. The proposed LPP-GO demonstrated significant improvements over LPP in terms of path length and travel time. In the frontal collision scenario, LPP-GO reduced the path length by approximately 8 % and shortened the travel time by 20 %, thereby reaching the target destination without collision. In the side collision scenario, although the path length increased slightly, the travel time decreased by 36 %, indicating more efficient avoidance without any collisions. In the multiple-obstacle scenario, the path length was reduced by >44 %, and the travel time was reduced by 64 %, proving its superior navigational capabilities in complex environments.

In particular, early on in multi-obstacle scenarios, the classic LPP would collide early because of trying to avoid *Obstacle 1*. These incidents, which occurred because path planning did not consider the

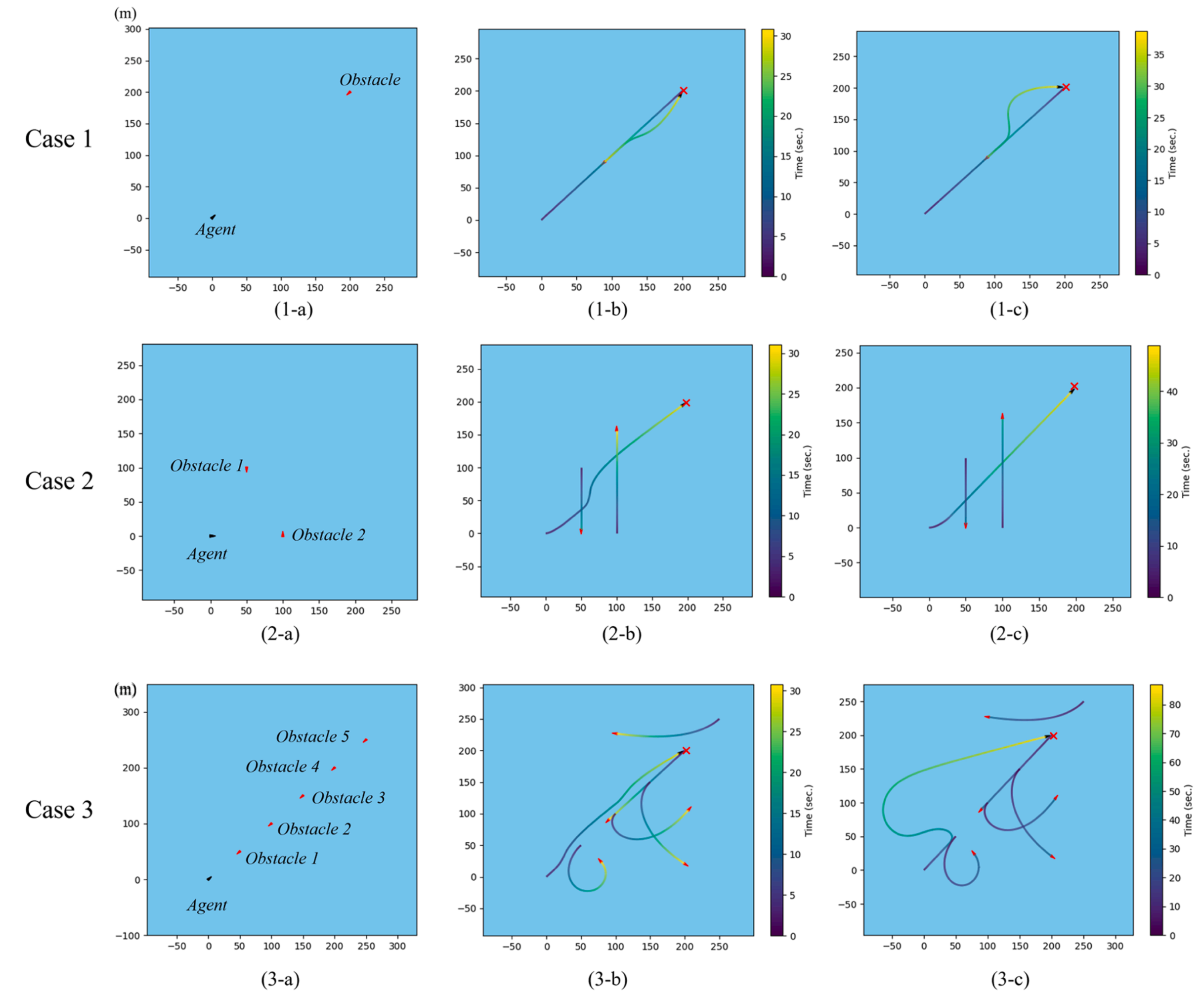

Fig. 8. Experiment 1(b) results. (1-a) Frontal collision scenario. (1-b) LPP-GO result. (1-c) Classic LPP result. (2-a) Side collision scenario. (2-b) LPP-GO result. (2-c) Classic LPP result. (3-a) Multi-obstacle collision scenario. (3-b) LPP-GO result. (3-c) Classic LPP result.

Table 5
Quantitative results of experiment 1.

		Case 1	Case 2	Case 3
LPP-GO	Trajectory length (m)	277.0	281.0	276.8
	Travel time (s)	31.2	31.4	31.1
	Number of collisions	0	0	0
Classic LPP	Trajectory length (m)	301.4	275.5	493.7
	Travel time (s)	39.1	49.3	87.3
	Number of collisions	0	0	1

speed of dynamic obstacles, highlight the ability of LPP-GO to effectively integrate dynamic obstacle movement and destination information to generate safe and efficient routes. Collision occurrence with classic LPP emphasizes the need for LPP-GO to respond more sensitively to complex dynamic changes. However, it stresses the importance of setting appropriate parameters in LPP-GO to prevent overly aggressive avoidance behaviors.

These results clearly demonstrate that the proposed LPP-GO outperforms the classic LPP by proactively handling obstacles, enabling the generation of safer and more efficient paths even in complex environments. Unlike traditional algorithms, which often rely on reactive adjustments after detecting obstacles, LPP-GO predicts and pre-emptively adjusts waypoints to minimize sudden detours or inefficient route changes.

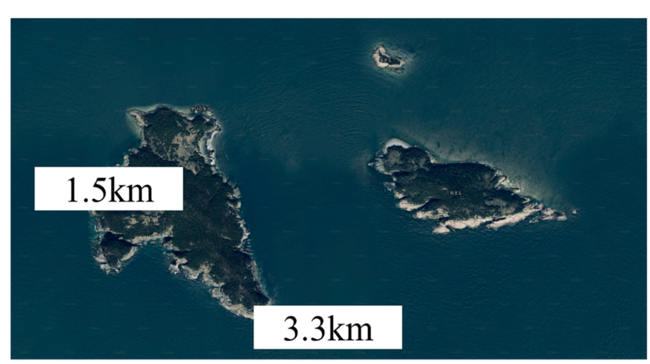

Fig. 9. Small-scale map of real-world scenario to validate proposed path planning framework.

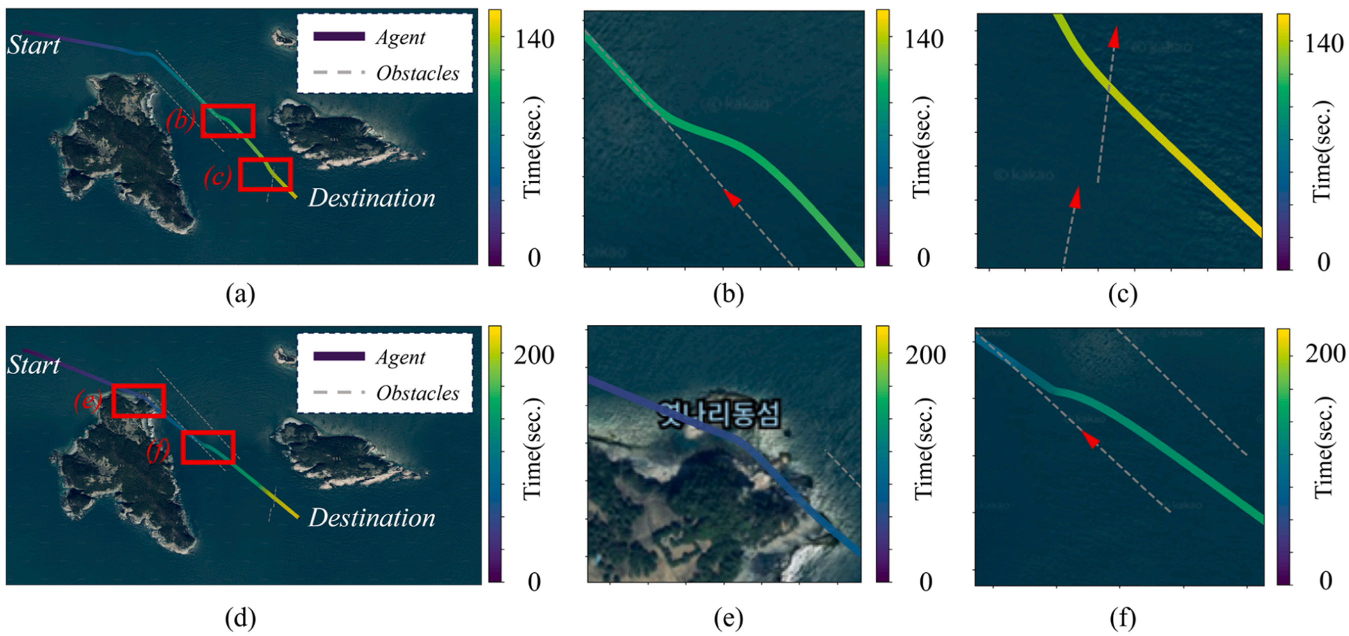

Fig. 10. Experiment 1(c) result. (a) Path from the proposed framework. (b) Initial collision avoidance in proposed framework (c) Final collision avoidance in proposed framework. (d) Path from the classic framework. (e) Collision in classic frameworks. (f) collision avoidance in classic framework .

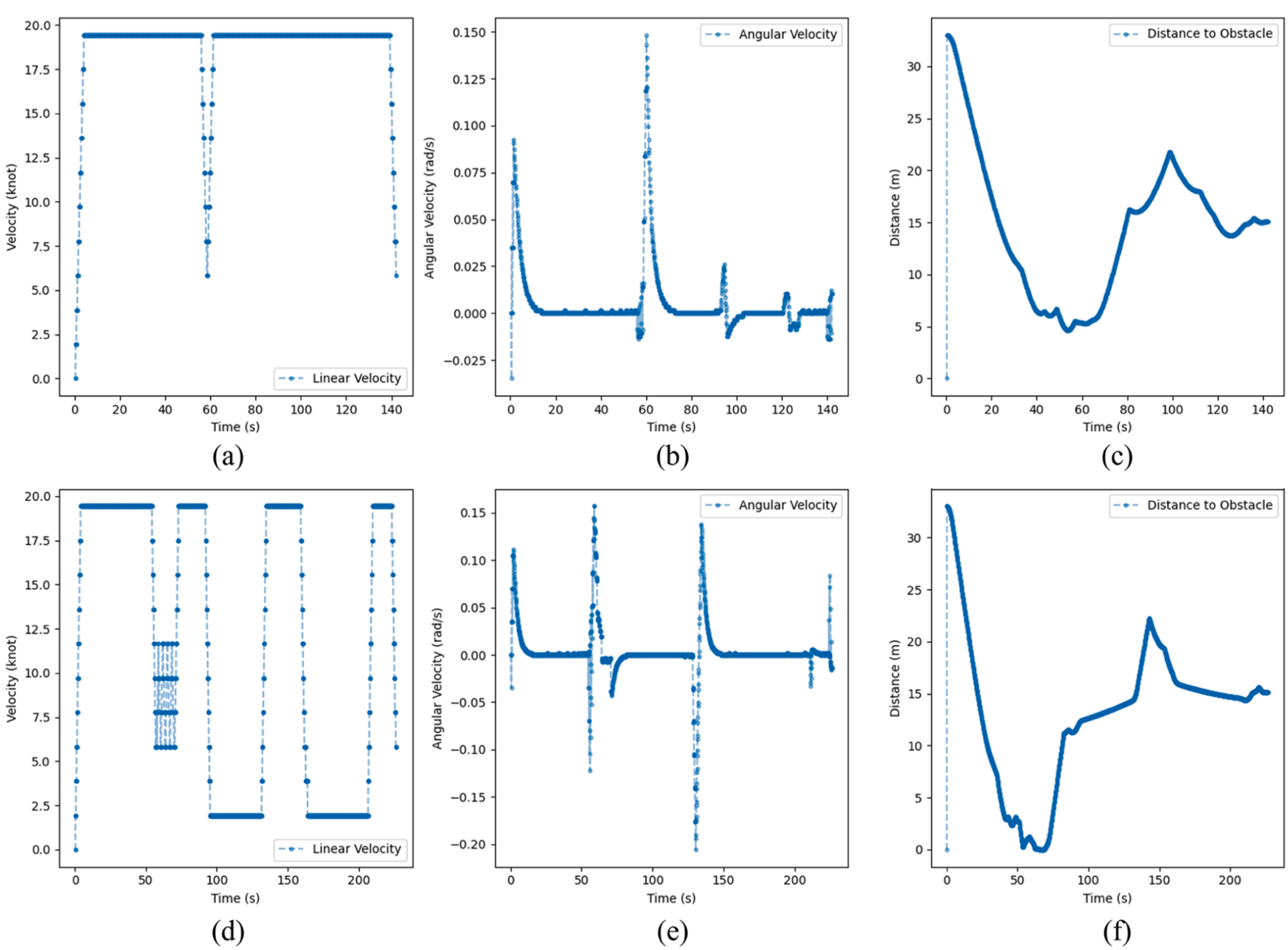

Fig. 11. Key parameters for path planning using the proposed and classic path planning framework. (a) Linear velocity of proposed framework. (b) Agular velocity of proposed framework. (c) Distance to obstacle of proposed framework. (d) Linear velocity of classic framework. (e) Agular velocity of classic framework. (f) Distance to obstacle of classic framework.

This proactive approach reduces the frequency of abrupt path recalculations and optimizes trajectory smoothness, resulting in lower energy consumption and faster navigation times. Additionally, by addressing potential collisions in advance, LPP-GO enhances overall system stability and operational reliability, which are critical in dynamic maritime scenarios.

4.2.3. Experiment 1(c): behavior validation using path planning framework

Fig. 9 presents an aerial image of the testing environment, illustrating the area with dimensions of 3.3 km by 1.5 km. In this complex maritime setting, integrated GPP and LPP were evaluated for their capability to interact with specified dynamic environmental elements and execute path planning. The experiment aimed to determine whether the USV could effectively navigate obstacles and efficiently reach its intended destination.

Fig. 10 presents the results of Experiment 1(c), showing the path planning results of the proposed and classic frameworks near a real island in South Korea. Figs. 10(a) and (d) show the overall paths taken by the proposed and classic frameworks, respectively. Figs. 10(b), (c), (e), and (f) highlight the specific moments of avoiding dynamic obstacles and explain how each framework approaches them.

Specifically, Figs. 10(b) and 10 (c) show that the proposed framework efficiently plans a path while avoiding dynamic obstacles and terrain. The proposed framework achieved a voyage time of approximately 140 s by choosing straight and short paths while maintaining a safe distance as it approached the obstacles. It also minimized unnecessary path modifications during the obstacle avoidance process, thereby improving the consistency and predictability of path planning.

Figs. 10(e) and 10 (f), which illustrate the classic framework, reveal that more complex and longer paths were generated during the obstacle avoidance process. These paths required frequent directional changes around obstacles, leading to an increased overall travel time of approximately 200 s. Particularly, as demonstrated in Figs. 10(e) and (f), the classic framework produced paths that collide with the terrain and create inefficient routes near the islands. These factors significantly degrade the safety and efficiency of navigation.

Fig. 11 provides a quantitative analysis of key path planning parameters using both the proposed and classic frameworks. This figure displays linear velocity (Fig. 11(a), 11 (d)), angular velocity (Fig. 11(b), 11 (e)), and distance from obstacles (Fig. 11(c), 11 (f)). These parameters are essential for evaluating the performance of the frameworks under dynamic conditions.

In the detailed analysis of the proposed framework, Fig. 11(a) shows that the linear velocity is generally stable, with strategic reductions in speed around the 60 and 140-second marks to safely navigate around obstacles. Fig. 11(b) indicates that major adjustments in angular velocity occur when avoiding obstacles, demonstrating the framework's ability to respond quickly and accurately to adjust the path. Furthermore, Fig. 11(c) shows that the distance to obstacles is consistently maintained between 5 and 30 meters, suggesting that the framework ensures a safe distance to prevent collisions with obstacles.

Conversely, the classic framework depicted in Figs. 11(d), 11 (e), and 11 (f) exhibits greater variability in these path planning parameters. Particularly, Fig. 11(d) shows an irregular pattern in linear velocity, with less frequent strategic speed adjustments, implying potential inefficacy in obstacle avoidance. Angular velocity changes in Fig. 11(e) also show sharp variations when encountering obstacles, which could lead to decreased stability due to less precise adjustments. Additionally, as illustrated in Fig. 11(f), the distance to obstacles fluctuates irregularly, with critical closeness to obstacles around the 100-second mark, potentially resulting in collisions.

Experiment 1 involved a comparative analysis between the proposed path planning framework and the classic method, focusing on their obstacle avoidance and safe path generation capabilities. The results quantitatively demonstrate that the proposed framework significantly outperforms the classic method in terms of maintaining safe distances, ensuring stable speeds, and facilitating smoother navigation adjustments. These improvements highlight the proposed framework's effectiveness in guaranteeing safe and efficient navigation even under complex maritime conditions.

This performance difference stems from the proactive obstacle-handling mechanism embedded in the proposed framework, which utilizes safety areas and an improved objective function. Unlike traditional approaches, which primarily rely on reactive adjustments after obstacle detection, the proposed method predicts and preemptively adjusts waypoints to avoid potential collisions using these safety areas and optimized parameters defined in the improved objective function. This predictive capability reduces abrupt path changes, minimizes computational overhead, and enhances overall navigation stability and efficiency. Additionally, the framework dynamically adjusts key operational parameters during path recalculations, guided by the improved objective function, allowing it to respond effectively to changing obstacle dynamics and environmental conditions.

4.3. Experiment 2

This section focuses on Experiment 2, which aimed to evaluate statistical performance. Fig. 12 shows a small 1.3 km x 1.1 km map designed specifically for this experiment. It contains starting, dynamic obstacles, and destination points. To increase the robustness of the test, the start and destination points were randomly generated for each simulation. Dynamic obstacle sectors were strategically placed in areas with a high crash risk. The setup evaluated the ability of the framework to effectively handle complex conditions and sudden changes.

Fig. 13 presents a comparison of the simulation results between the proposed and classic path planning frameworks around an island based on simulations conducted on the small-scale real-world map shown in Fig. 13. In Fig. 13(a), the proposed path planning framework demonstrates a systematic and focused navigation strategy. The paths were fewer and exhibited a clear and orderly pattern, indicating a predictable and reliable planning process. This method not only maintained safer distances from the island but also significantly reduced the time needed to complete the paths, with most taking <100 s, as shown by the heatmap of travel time. This efficiency emphasizes the improved performance of the proposed framework in managing path planning tasks.

Conversely, Fig. 13(b) shows the classic path planning framework, which generated a wide array of paths suggesting inconsistencies and potential inefficiencies in the planning approach. Some paths veer dangerously close to the island, raising concerns about possible

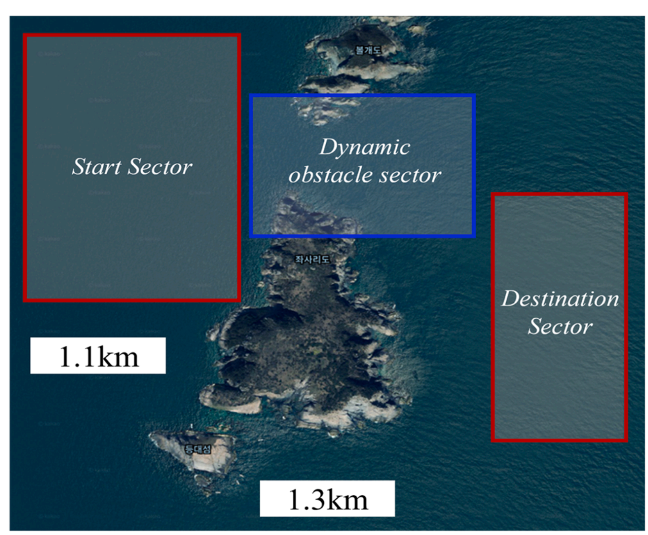

Fig. 13. Experiment 2 result. (a) Results using the classic path planning framework. (b) Results using the proposed path planning framework.

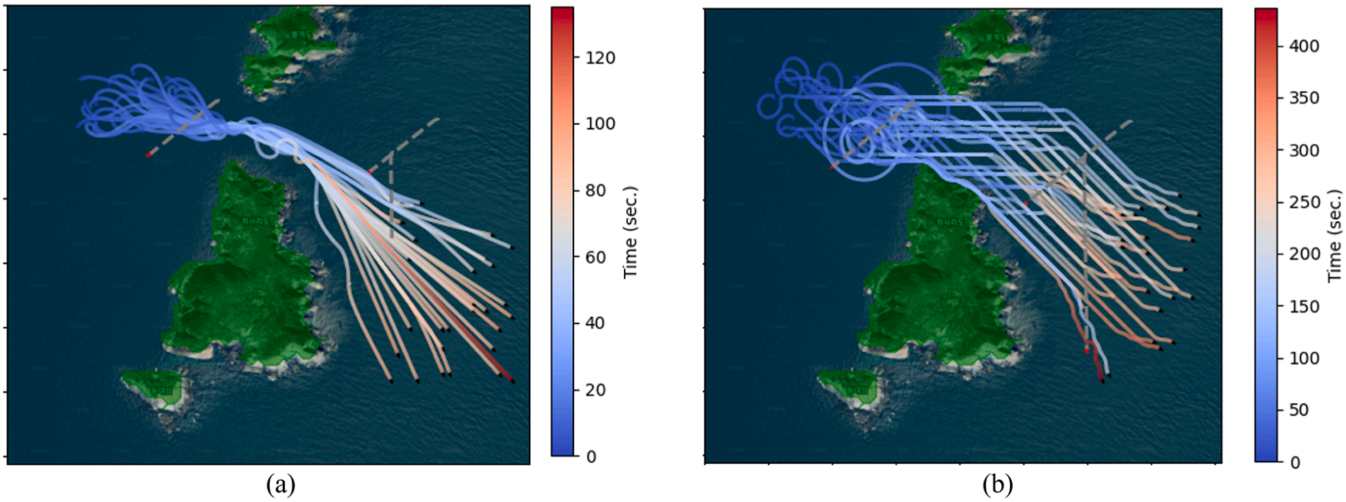

Fig. 12. Small-scale real-world map to statistically validate the proposed path planning framework.

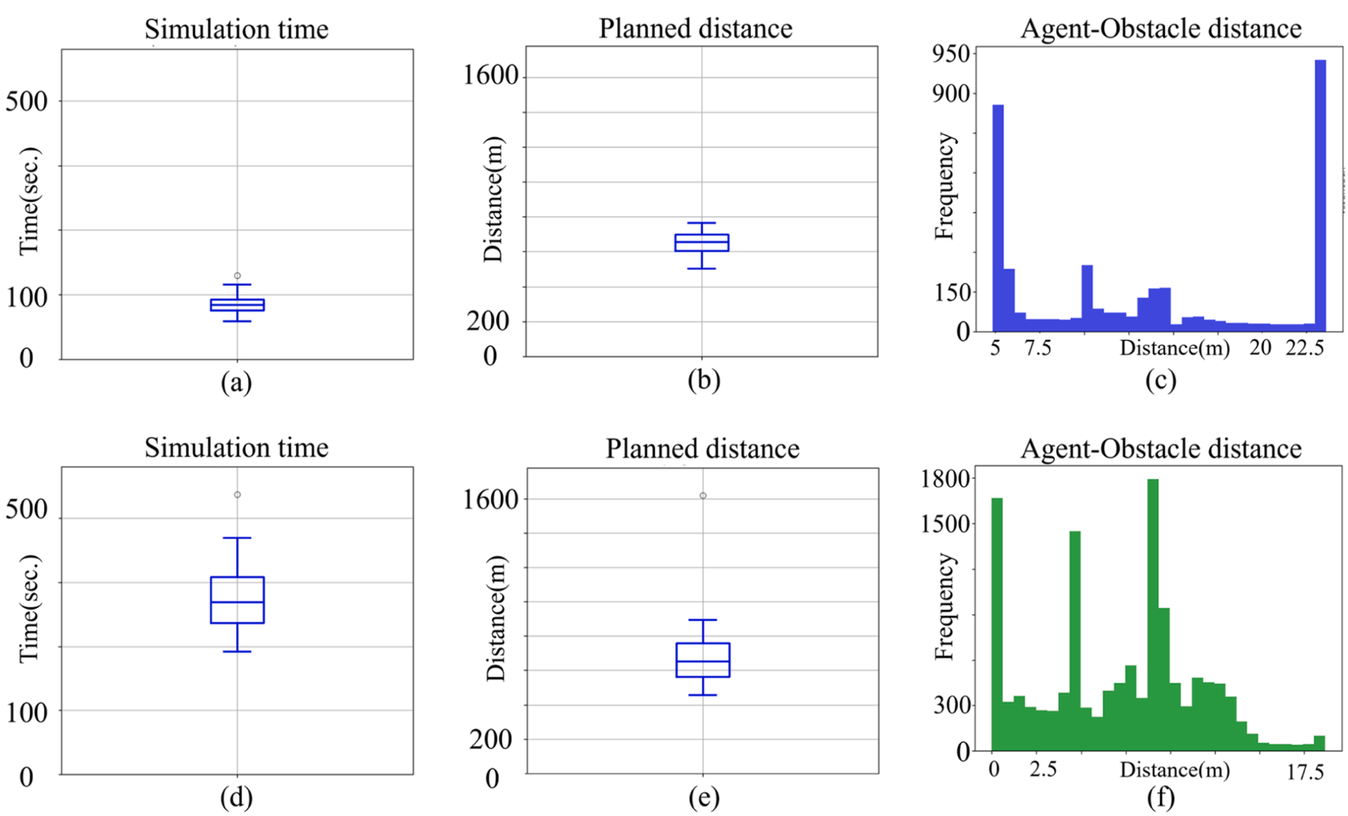

Fig. 14. Experiment 2 statistical simulation results. (a) Simulation time of proposed framework (b) Path length of proposed framework. (c) Agent-obstacle dist. of classic framework. (d) Simulation time of classic framework. (e) Path length of classic framework. (f) Agent-obstacle dist. of classic framework.

navigation errors and risks of collision. In addition, some routes were generated that did not converge on the desired destination and continued to circle around. The heatmap of travel time analysis shows that many of these paths required over 400 s to complete, highlighting the inefficiency in generating effective routes.

Fig. 14 quantitatively compares the performance differences between the classic and proposed path planning frameworks. Sub-figures (a, b, c) relate to the proposed framework, whereas subsequent sub-figures (d, e, f) correspond to the classic framework. This layout provides a clear basis for comparing the two approaches in terms of the simulation time, path length, and distance from obstacles.

The proposed framework shown in Fig. 14(a) processed the simulation in <100 s, which is significantly faster than the 500 s required by the classic framework as shown in Fig. 14(d). The reduction in the simulation processing highlights the efficiency of the proposed system and its ability to react quickly in dynamic environments. This improvement can be largely attributed to the proactive obstacle-handling mechanism, which allows the system to anticipate obstacles in advance and plan paths with potential collisions already considered. By incorporating anticipated obstacle locations into the path planning process, the framework minimizes abrupt adjustments during execution and ensures smoother navigation.

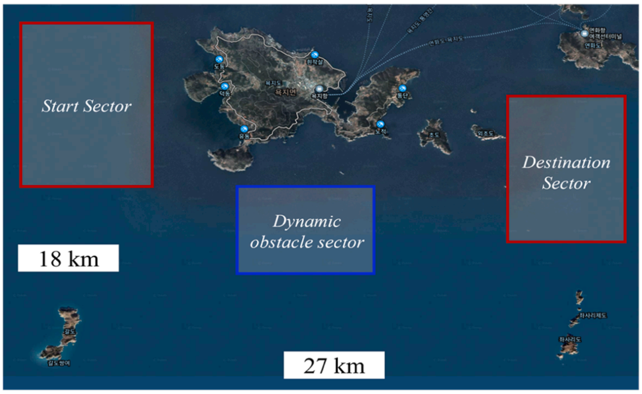

Fig. 15. Large-scale real-world map to statistically validate the proposed path planning framework.

Figs. 14(c) and 14 (f) display histograms representing the distribution of distances between the agents and nearest obstacles during the simulations, illustrating the collision avoidance capabilities of each framework. In Fig. 14(c), which represents the proposed framework, there are no instances where the measured distance reaches zero, indicating that no collisions were recorded owing to the effective avoidance strategies. The closest approaches consistently exceeded a safe threshold, demonstrating that the framework reliably maintained a safe distance from obstacles, thereby minimizing the risk of collisions.

Conversely, Fig. 14(f), representing the classic framework, shows several instances in which the measured distances are zero, clearly indicating the occurrence of collisions owing to ineffective navigation around obstacles. This clear difference in the minimum distances emphasizes the superior performance of the proposed framework in ensuring safety and reducing the likelihood of navigation errors in complex environments.

In summary, although the path lengths shown in Figs. 14(b) and (e) are similar, there is a significant difference in the simulation times. The classic framework averaged approximately 500 s, whereas the proposed framework processed <100 s, achieving an approximately 80 % reduction in time. This efficiency in path planning significantly enhances the ability to respond quickly in dynamic environments, offering substantial advantages in tasks where a rapid response is crucial. These quantitative analyses are vital for objectively demonstrating the superior performance of the proposed path planning framework and showcasing its capability to enable safe and efficient navigation in complex maritime environments.

4.4. Experiment 3

In this section, we analyze the results of Experiment 3. Fig. 15 shows the large-scale real-world map measuring 27 × 18 km used in Experiment 3. Like Experiment 2, this setup incorporated a start sector, a dynamic obstacle sector, and a destination sector, but on a much larger scale. As in Experiment 2, both the starting point and destination were generated at random coordinates in each simulation and the dynamic obstacle sector was strategically placed in areas of high collision risk. This large-scale real-world environment focused on evaluating the capabilities of the framework to navigate extensive and complex conditions and adapt to sudden changes.

Fig. 16 shows the results of iterative simulations on a large-scale real-world map, comparing the performance of the proposed path planning framework and the existing framework. As shown in Fig. 16(a), the proposed framework generated more direct and shorter paths than the existing framework, significantly reducing the travel time represented by the heatmap, where most of the paths were completed within 4000 s. The visual demonstration highlights the efficiency and effectiveness of the proposed system and emphasizes its capability for a fast response in dynamic environments.

By contrast, Fig. 16(b) shows that the classical framework generated more distributed and indirect paths, often with travel times exceeding 8000 s, demonstrating inefficiency and potential delays. This inefficiency primarily stems from the classic framework's reactive approach to obstacle handling, where obstacles are addressed only after detection rather than being anticipated in advance. As a result, the system frequently produces zigzag paths, increasing both travel time and energy consumption.

However, the proposed framework consistently maintained safe distances from obstacles, leading to smoother and safer navigation and significantly enhancing time efficiency and navigational safety over classic methods. Despite its strengths, the proposed framework occasionally struggled in situations requiring significant maneuvering, failing to extract sufficient waypoints and resulting in less efficient routes, highlighting areas that need further improvement.

Fig. 17 quantitatively compares the performance differences between the classic and proposed path planning frameworks on a large-scale map. Sub-figures (a), (b), and (c) depict the results of the proposed framework, whereas (d), (e), and (f) correspond to the classic framework. As described in Fig. 14, this arrangement provides a clear basis for comparing both approaches in terms of simulation time, path length, and distance from obstacles, further emphasizing the improvements in efficiency and safety offered by the proposed framework in navigating large and complex environments.

In terms of the simulation time, Figs. 17(a) and 18 (d) distinctly

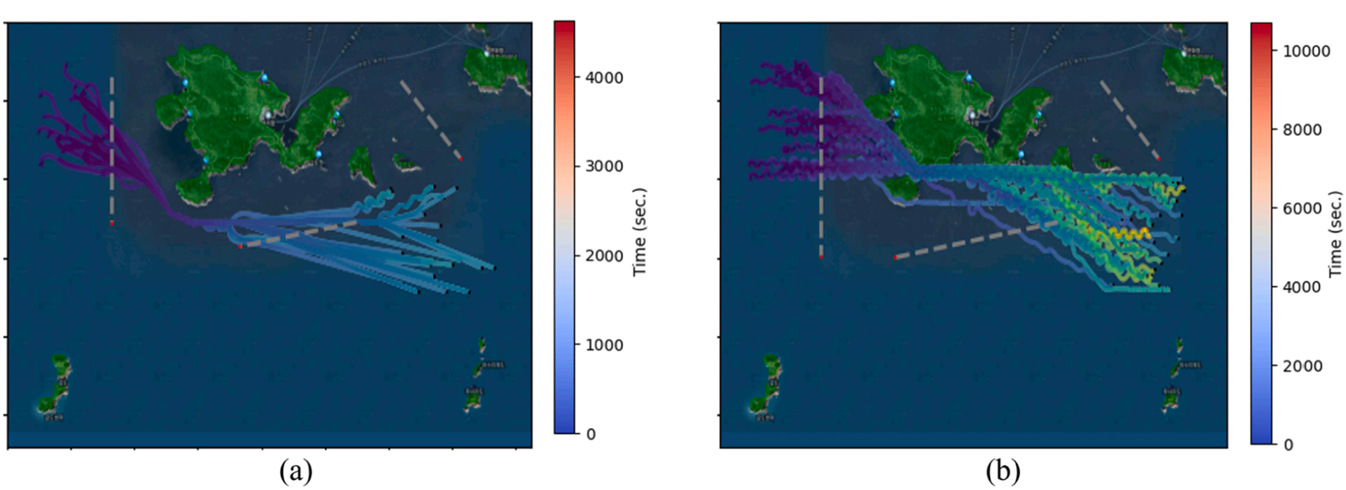

Fig. 16. Experiment 3 statistical simulation results. (a) Proposed path planning framework. (b) Classic path planning framework.

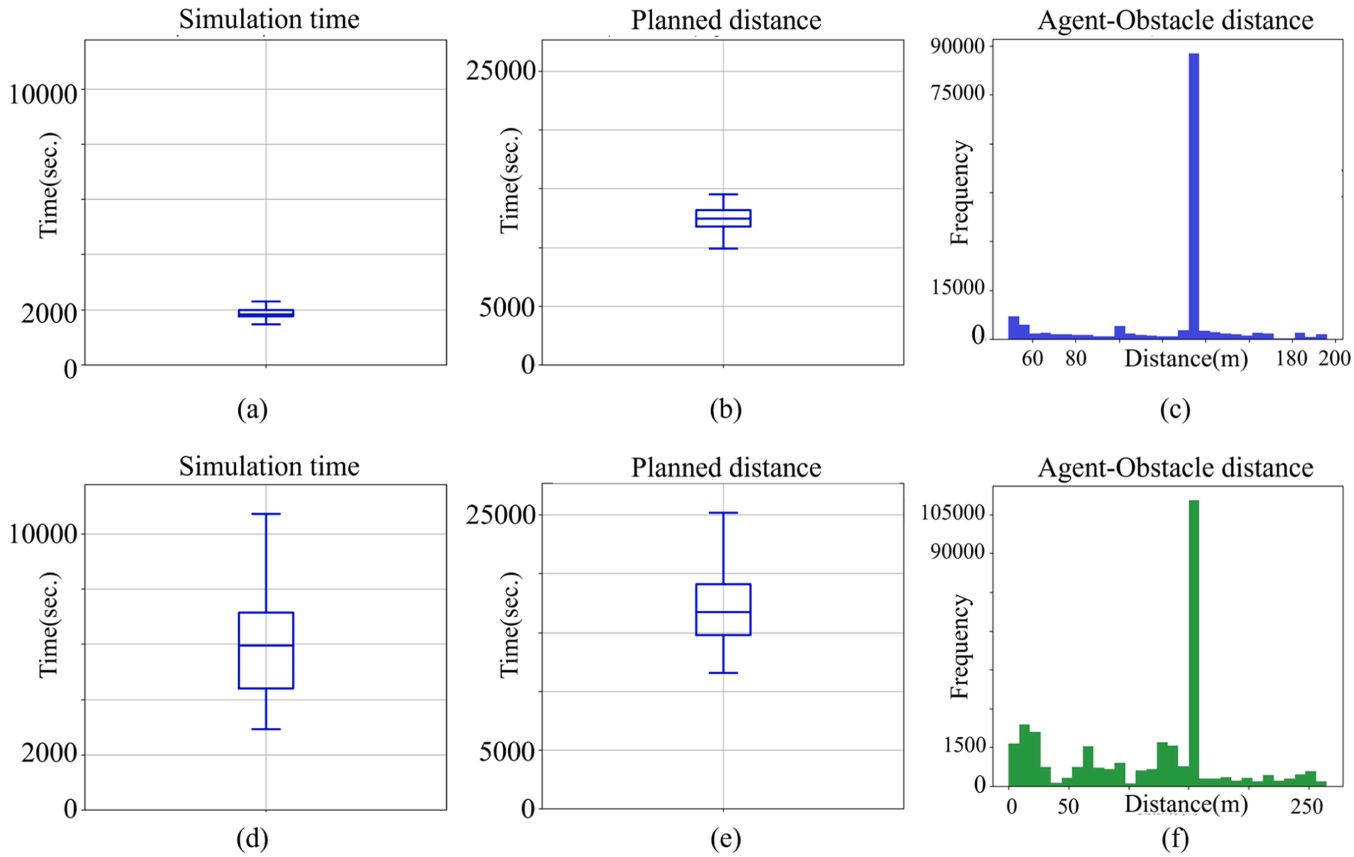

Fig. 17. Statistical simulation results for large scale map. (a) Simulation time of proposed framework (b) Path length of proposed framework. (c) Agent-obstacle dist. of classic framework. (d) Simulation time of classic framework. (e) Path length of classic framework. (f) Agent-obstacle dist. of classic framework.

highlight the efficiency differences between the two frameworks. Fig. 17 (a) demonstrates an average simulation time of approximately 2000 s, which is significantly less than over 6000 s, as shown in Fig. 17(d). This considerable reduction underlines the ability of the proposed framework to quickly optimize paths and enhance efficiency in complex environments.

In terms of path length, the two frameworks produced similar overall distances, as demonstrated by the statistical analysis. However, the proposed framework maintained a much more consistent path length distribution, emphasizing its ability to plan paths more accurately and efficiently. In particular, the statistical results show that the average lengths of the planned distances are similar, but the proposed framework maintains a more accurate estimated traveled distance across multiple simulations, owing to reduced variability.

Furthermore, the distribution of the distance between the agent and nearest obstacle is consistent with the proposed framework, maintaining a distance of at least 30 m. Consequently, it effectively avoided collisions with obstacles. The consistent maintenance of safe distances is reflected in the histograms of the simulations. It enhances the safety and reliability of navigation, particularly in environments with dynamic obstacles where maintaining a safe distance is important for preventing potential accidents.

Conversely, the classic framework shown in Fig. 17(f) frequently recorded distances reaching zero meters, clearly indicating the risk of collisions and frequent close encounters with obstacles. These results illustrate the superior obstacle avoidance capabilities of the proposed framework and demonstrate that it enables safe navigation in complex environments compared to classical methods.

4.5. Discussion

The proposed path planning framework was validated through simulation-based experiments designed to replicate complex maritime navigation scenarios, including dynamic obstacles and varying operational conditions. The experimental results demonstrated that the framework outperforms existing algorithms by generating safer and more efficient paths, dynamically adapting to changing obstacles, and maintaining consistent performance across diverse simulation environments. However, it must be acknowledged that this framework has not yet been tested in real-world maritime environments. While simulation scenarios are valuable for initial validation, they cannot fully replicate sensor inaccuracies, environmental variability, and other real-world constraints. This limitation raises uncertainties regarding the framework's performance when deployed in practical maritime scenarios.

If the proposed path planning framework were deployed in an actual maritime environment, discrepancies between simulation results and actual performance, commonly referred to as the gap between simulation and reality, would inevitably arise. This gap is primarily due to two factors: sensor noise and dynamic environmental variability. While simulation provides stable and noise free conditions, real world environments introduce unpredictable variables that cannot be fully captured in predetermined scenarios.

Sensor noise is a major contributor to this gap. Sensors such as lidar, radar and cameras are highly sensitive to vibrations caused by waves, sea spray and varying lighting conditions. These factors distort sensor data, leading to inaccurate obstacle detection, false alarms and reduced situational awareness. As a result, path planning algorithms may make suboptimal decisions, resulting in inefficient or unsafe navigation. Furthermore, sensor calibration errors in real world deployments can exacerbate these inaccuracies and further reduce the reliability of the system in dynamic conditions.

Dynamic environmental variability represents another significant challenge. Maritime environments are inherently unstable, characterized by fluctuating wave patterns, sudden changes in wind direction and shifting ocean currents. These dynamic factors continuously alter the operational context, rendering the static assumptions used in numerical models unreliable. For example, a trajectory optimized for calm waters may become infeasible when unexpected high waves or strong currents occur. Such unpredictable changes can lead to path deviations, increased energy consumption and delays in reaching target destinations.

These challenges expose the fundamental limitations of static numerical modeling approaches that rely on predefined parameters and assumptions that cannot adapt to the fluid nature of real-world marine environments. Addressing this gap requires a fundamental transition to data-driven adaptation mechanisms that are capable of real-time adjustments based on real-world sensor feedback and dynamic environmental cues. Without such mechanisms, frameworks will face significant challenges in ensuring reliable, safe, and efficient path planning during real-world deployments.

5. Conclusion

In this study, we propose an advanced path planning framework for the navigation of USVs in marine environments. The main contribution of this study is the combination of GPP-SA and LPP-GO to enable reliable and efficient navigation of USVs in complex marine environments. GPP-SA enhances stability through path planning considering the marine terrain and safety zones, whereas LPP-GO utilizes predictions of the movement of target points and dynamic obstacles to plan the optimal path. Additionally, the framework integrates proactive obstacle handling mechanisms that leverage safety areas and an improved objective function to minimize abrupt heading changes and sudden speed adjustments, ensuring smooth trajectory planning and enhanced system stability. To validate the performance of the proposed framework, we conducted simulations with real ship parameters and found that it performed well, especially in dynamic environments, with an approximately 15 % reduction in path length and a 60 % reduction in travel time compared to conventional methods.

The proposed path planning framework in this paper has been validated through various simulations, yet it still exhibits several limitations. Notably, the simulation models employed did not incorporate complex dynamical factors such as ocean currents and wave actions, which are critical in influencing the navigation of USVs in actual marine environments. Additionally, the framework was not tested on actual hardware in real-world settings, which limits our ability to identify and address potential issues that may arise during operational deployment.

Furthermore, the developed LPP-GO algorithm in this study focused on predicting the linear movements of dynamic obstacles. However, obstacle movements in real-world settings can often be non-linear, necessitating more advanced modeling techniques for effective response. To improve this, future research will explore the use of deep learning-based algorithms to predict and adapt to both linear and nonlinear movements of dynamic obstacles. This approach is expected to enhance the accuracy and reliability of the path planning process significantly.

In conclusion, the proposed framework integrates multiple advanced mechanisms, including safety areas, an improved objective function, and predictive waypoint adjustments, to address the key challenges in marine navigation environments. While the simulation results demonstrate promising outcomes, future efforts will focus on bridging the Sim-to-Real gap, incorporating complex environmental dynamics, and refining the algorithm's adaptability to real-world operational scenarios. These improvements will be crucial for advancing the robustness and applicability of the framework in practical USV operations.

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Ethical approval

This study did not involve any human participants, human data, or human tissue. All applicable international, national, and/or institutional guidelines for the care and use of animals were followed.

This statement is formulated in accordance with the submission guidelines of Applied Ocean Research and includes all the necessary declarations.

CRediT authorship contribution statement

Hee-Mun Park: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Seung-Wan Cho:** Writing – original draft, Validation, Software. **Kyung-Min Seo:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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